

RESEARCH ARTICLE

FedHealthcare: Federated Learning and Lightweight Additive Homomorphic Encryption-Based Privacy-Preserving Healthcare

Tapasi Bhattacharjee¹  | Anwarul Haque¹ | Faisal Shamim¹ | Debashis De²

¹Department of Information Technology, Techno International New Town, Kolkata, India | ²Department of Computer Science and Engineering, Maulana Abul Kalam Azad University of Technology, Nadia, India

Correspondence: Tapasi Bhattacharjee (dr.tapasi.bhattacharjee@tint.edu.in)

Received: 11 June 2025 | **Revised:** 18 September 2025 | **Accepted:** 24 September 2025

Funding: The authors received no specific funding for this work.

Keywords: cryptosystems | electronic health records | federated learning | healthcare data | homomorphic encryption | machine learning | privacy-preserving

ABSTRACT

In recent days, collaborative health data analysis is conducted in various health organizations. Hence, data privacy and security are major concerns for healthcare industries. The existence of strict regulations underscores the urgent need for secure and compliant data-sharing solutions. To that aim, this paper proposes FedHealthcare, a privacy-preserving machine learning (ML) framework that integrates federated learning (FL) with lightweight additive homomorphic encryption (HE). This scheme allows every healthcare organization to train a local model, and it uses lightweight additive HE to encrypt the sensitive parameters. After every round, all clients receive the encrypted updates that have been safely combined on a global server via homomorphic addition. This conceals the raw data. Compressed gradient aggregation and adaptive encryption preserve high accuracy and privacy rules while consuming less bandwidth and computation. Not only does it encrypt the sensitive model parameters, but it also integrates the compressed gradient aggregation. This improves training efficiency without compromising accuracy. Experiments are conducted on realistic healthcare datasets. An accuracy achievement of more than 90.8% is possible using FedHealthcare with lower bandwidth usage (250 KB/round) and a 20% improvement in encryption speed compared to full HE approaches. The improved results demonstrate that the integration of FL and HE works well to protect privacy while preserving high model performance. This makes FedHealthcare a good option for extensive medical AI applications.

1 | Introduction

Healthcare industries are producing an enormous volume of useful patient data. This data plays an important role in various sectors. Data from electronic medical records (EMR) and sophisticated diagnostic systems are essential for enhancing patient outcomes and advancing innovative research [1–4]. Concerns

regarding security and privacy, however, are growing along with the dependence on data. While institutions must adhere to stringent standards like the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in Europe, handling sensitive information is a big concern for the patients [5–8]. Hence, investigating privacy-preserving techniques for healthcare data is quite

Tapasi Bhattacharjee, Anwarul Haque, Faisal Shamim, and Debashis De contributed equally to this study.

crucial. There are various issues in sharing the personal data of the patients [9–11].

There are some requirements for strict security measures due to the nature of healthcare data as these data are closely linked to human life, health, and privacy. Health data sharing between institutions may be affected by privacy and security issues [12–15]. As a result, datasets are fragmented and model performance is becoming poor. Global optimization is frequently not achieved by models that are trained separately by the institutions. Therefore, new approaches are required to strike a compromise between the potential of collaborative machine learning (ML) and the privacy requirements of healthcare data [16–21].

The importance of secure authentication protocols in healthcare-oriented Internet of Things (IoT) environments is being studied in recent years. A robust Radio Frequency IDentification (RFID) authentication protocol based on the digital Schnorr cryptosystem is proposed by Alazab et al. [22]. This scheme is based on COVID-19 healthcare scenarios. This scheme ensures the reliability and security for IoT-enabled medical systems. This scheme addresses the issues faced by lightweight encryption methodologies and secure device communication in constrained environments. Later on, Shariq et al. [23] proposed a lightweight and secure RFID-based protocol for healthcare systems. It guarantees security and privacy characteristics, lowers the communication costs, and requires fewer computational operations. However, implementing this scheme for real-life applications is considered the scope of future work. Later on, an efficient and reliable ultra-lightweight RFID authentication scheme is reported in [24]. This scheme enhances patients' medication safety. It provides a lower computation and storage cost for RFID tags; hence, it is feasible to apply in real-time healthcare applications.

A decentralized method called federated learning (FL) enables several data holders to work together to train models without exchanging the raw data. Institutions exclusively share aggregated updates instead, guaranteeing local control over data [13, 15, 25–26]. It is worth mentioning that anonymity plays a

very crucial role in the healthcare industries. Data aggregation is extremely advantageous. FL reduces the risks of breaches by enabling ML advancements. Here, most of the frameworks are dependent on central aggregation to coordinate updates. Hence, the risk of a single point of failure is an important challenge for FL applications. If this server fails or is compromised, the entire training process can be disrupted [26]. However, few decentralized strategies may be used to address these issues. These strategies include blockchain-enabled smart contracts, distributed aggregators, and so on. These methodologies are advantageous in reducing the central dependency, improving robustness, and ensuring trust in healthcare organizations [14, 21]. The integration of FL with homomorphic encryption (HE) is advantageous in improving the privacy of data. It enables calculations on encrypted data without any requirement of decryption. For safe aggregation, health organizations can train the models locally. These are then encrypted and transmitted to a central server. In recent days, the amount of healthcare data is increasing due to the advancements of IoT and electronic health records (EHRs). Hence, privacy preserving is becoming crucial [6, 8, 9, 27–28]. On the one hand, HE assures ongoing data confidentiality; on the other hand, FL ensures collaborative learning. This combination paves the way for secure predictive analytics, improving patient outcomes and advancing medical research. The diagrammatic illustration of work-flow in the integration of FL and HE for healthcare organizations is represented in Figure 1.

Recent advancements in privacy-preserving FL have explored the integration with HE to improve the security in distributed ML environments. Antunes et al. [29] proposed a general architecture for FL applied to healthcare data based. It explores improvements the aggregation mechanisms required to generate the learning model from distributed contributions. Later on, Jin et al. [30] proposed FedML-HE, an efficient FL system that incorporates HE to secure model updates while reducing the computational burden typically associated with full HE schemes. This scheme emphasizes scalability and modular design. Practical deployment in various collaborative learning scenarios is possible using this scheme.

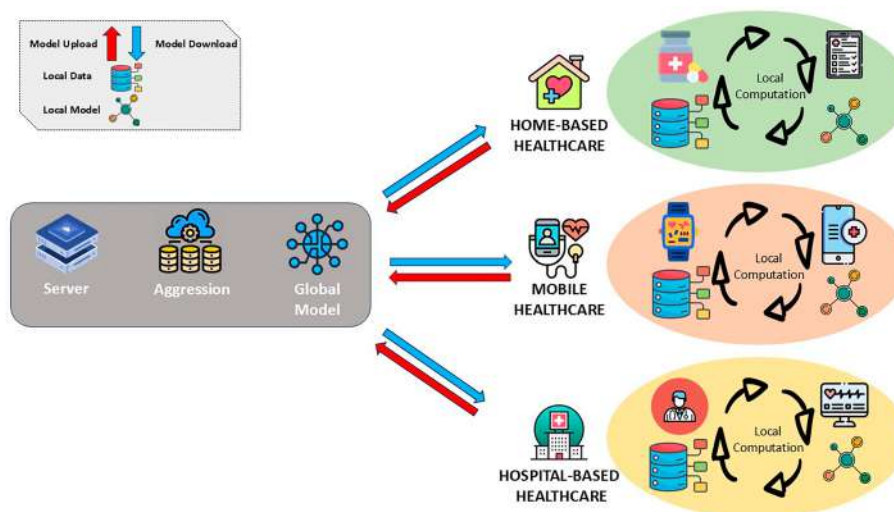


FIGURE 1 | Privacy preservation using FL and HE in health organizations.

Most of the integrated schemes reported so far suffer from high overhead and are computationally complex. Hence, designing a lightweight HE approach with less computational burden deserves further study. To this aim, this study proposes an integrated framework of FL with lightweight additive HE, named as “FedHealthcare.” This scheme enhances data privacy in healthcare applications. FL enables model training across decentralized data sources, allowing collaboration among multiple institutions without centralizing data. At the same time, HE offers a method for securely aggregating encrypted model updates from participating entities, such as hospitals. In this scheme, lightweight additive HE encrypts only the essential model parameters. It reduces the computational burden while preserving the privacy of health data. The integration of FL and HE effectively supports the collaborative training of ML models while preserving data confidentiality. FedHealthcare shows significant promise in accurately predicting patient-related outcomes. These include readmission rates and treatment efficacy while maintaining the security of sensitive information throughout the process. This study provides a scalable and compliant solution that enables secure collaboration among health organizations. Hence, it improves privacy-preserving analytics in the healthcare industry.

The rest of the paper is organized as follows: Section 2 presents some recently reported related works and the scope of the current work. The preliminary part is discussed in Section 3. Section 4 presents the proposed scheme. The results and performance evaluation are included in Section 5. In Section 6, the conclusions are drawn along with some scope for future works.

2 | Literature Review and Scope of Work

This section briefly presents a review of recently reported privacy-preserving frameworks on healthcare data. This discussion highlights the specific merits and limitations of these works. The scope and contributions of the proposed work are then reported in the next sub-section.

2.1 | Related Works

In the existing healthcare systems, there exist multiple issues that usually prevent information exchange among the healthcare institutions and stakeholders. These issues affect seamless collaborations too [19]. Secure data analysis in healthcare domains is being revolutionized by privacy-preserving technologies using FL and HE. HE performs computations on encrypted data, guaranteeing privacy even in the case of a security compromise. FL allows organizations to train models jointly without disclosing the raw data [4]. Hence, the integration of HE and FL presents a viable approach to privacy-preserving data analysis in the medical field. It allows several organizations to work together to train ML models without exchanging private medical information. It is compliant with laws such as GDPR and HIPAA [29]. A FL architecture supporting fully HE for privacy-preserving biomedical data analysis is reported in [18]. The performance of both non-encrypted and encrypted federated models was evaluated in this study. It shows how encrypted model updates can successfully defend against membership inference and gradient

inversion attacks. Hence it ensures secure learning without sacrificing model accuracy. However, this scheme lacks providing better security on the sensitive data. Stripelis et al. [10] utilized HE to secure all gradient information in FL. It stores the data on the cloud server to address the issue of data leakage in large-scale brain MRI datasets. This scheme demonstrated that the learning performance of the federated model remains consistent before and after encryption. Likewise, Ku et al. [31] addressed privacy concerns in medical data gathered by IoT devices through the use of HE. In their approach, data is encrypted and uploaded to a fog node, where model training is jointly carried out by the fog node and cloud server, enhancing the scheme's robustness and mitigating the high computational and storage demands of encrypted data.

A privacy-preserving FL framework called PPFLHE, based on was proposed by Wang et al. [11]. This scheme ensures the privacy and security of healthcare data. High classification accuracy and strong privacy preservation are achieved in this scheme. However, addressing communication overhead and improving training consistency through advanced privacy-preserving technologies are kept as their scope of future study. An efficient and privacy-preserving online diagnosis scheme for e-healthcare systems based on the FL mechanism was proposed in [2]. Here, the data is effectively protected by using FL along with the injection of blinding factors. Moreover, the physiological reports are efficiently classified through the employment of the support vector machine algorithm and a homomorphic cryptosystem. It ensures the protection of personal healthcare data. Security analysis and performance evaluations were conducted, demonstrating that the proposed scheme satisfies security requirements and achieves high diagnostic accuracy in real-world e-healthcare applications. Increasing the global model accuracy and robustness was considered as the scope of future work.

2.2 | Scope of FedHealthcare

In FedHealthcare, FL is used, where health organizations serve as individual clients. They use patient data and train ML models locally without sharing raw data. The privacy of the data is ensured by the integrated platform with HE. The three main modules of the architecture are health organizations as clients, the encryption, and the global server. To maintain the privacy laws such as HIPAA and GDPR, model training is done locally in health organizations. These are then encrypted, and the encrypted updates are only shared. The central server shares a global model with all the participating organizations. Each organization performs a public key encryption and changes it as per the local training. The global model may be improved by aggregating the encrypted updates rather than processing the raw data. In this method, there is no option for direct data transmission. Hence, data privacy is maintained, and model accuracy is also increased between the health organizations. Hence, to develop AI-driven healthcare solutions, FedHealthcare is advantageous and encourages collaboration across health organizations.

The proposed work focuses on recent advancements in FL with HE for healthcare applications. Few prior approaches enhanced security. But they also introduced considerable communication overhead [11]. The incorporation of edge computing helped

reduce this overhead, but requires additional infrastructure. Hence, it creates its own set of challenges. It is worth mentioning that data is not only plentiful, but also extremely valuable in the healthcare industry. Hence, the proper analysis of health data is helpful for patient care from diagnosis of disease to treatment of outcome prediction. There are various issues that can hamper data sharing and collaboration. In health organizations, maintaining security and privacy is very crucial. Most of the integrated FL-HE schemes reported so far encrypt all parameters or gradients. Hence, these schemes suffer from high overhead [2, 11].

Unlike traditional HE-based FL schemes, FedHealthcare introduces adaptive encryption, targeting only high-sensitivity model layers. This significantly reduces encryption overhead. The lightweight additive masking is used instead of expensive cryptosystems. In addition, gradient compression and client-side privacy verification improve communication efficiency and trust. This integrated design achieves over 90.8% accuracy with lower bandwidth usage (250 KB/round) and a 20% improvement in encryption speed compared to full HE approaches.

A lightweight adaptive encryption technique is used in the proposed scheme. It maximizes efficiency without sacrificing privacy by securing the relevant parameters only. Resilience against adversarial attacks and model inversion is improved by the integration of Differential Privacy (DP) and HE. Furthermore, in contrast to other cipher-text-based methods, the proposed FedHealthcare model compresses the encrypted updates. It speeds up the model convergence and uses less bandwidth. FedHealthcare guarantees adherence to privacy laws such as GDPR and HIPAA. These innovations position our research as a significant step forward in privacy-preserving collaborative learning for healthcare analytics. Figure 2 summarizes the related works [2, 11, 16, 18], key challenges in privacy-preserving FL and the unique contributions of our methodology in addressing these issues.

To explore the above-mentioned issues, the proposed methodology integrates HE and FL, which outperforms the current state-of-the-art research in five key areas. Each of them is backed by theoretical underpinning, mathematical modeling, and comparative analysis.

- **Optimized encryption efficiency**—FedHealthcare uses a lightweight additive HE scheme that encrypts only essential model parameters. Hence, it significantly reduces the computational and communication overhead. The equation for simplified additive HE is written in Equation (1).

$$c = E(m, P_k) = m + k \quad \text{or} \quad E(w_i^t) = w_i^t + k \quad (1)$$

where c is the encrypted message; $E(m, P_k)$ is the encryption function; m is the original model parameter; p_k is the public key; $E(w_i^t)$ is the encrypted weight; w_i^t is the local model weight; and k is the random mask.

Traditional public-key HE scheme like Paillier cryptosystem is computationally intensive. The proposed masked additive model achieves faster encryption and decryption, lower bandwidth, and additive HE properties without full-blown HE cost and is sufficient for secure aggregation in FL.

- **Selective/adaptive encryption strategy**—Inspired by model inversion attacks that target shallow layers, this scheme applies encryption to only sensitive layers or gradients. It reduces the overall HE computation cost. Let, $\theta = (\theta_1, \theta_2, \dots, \theta_n)$ be the model parameters. The encrypted parameters are θ_s (where, $\theta_s \in \theta$).

$$\theta_s = \theta_i \quad | \quad \text{high sensitivity (close to raw input)} \quad (2)$$

This leads to the time complexity of encryption, $O(\theta_s) \ll O(\theta)$

- **Defense against model inversion and poisoning attacks**—The proposed approach saves resources while maintaining security, proven effective against layer-wise attacks (gradient inversion attacks and model poisoning attacks).
- **Compressed gradient aggregation**—In the scheme, encrypted model updates are used with FedAvg aggregation applied to compressed gradients. If g_i^t be the gradient, then,

$$\begin{aligned} Q(g_i^t) &= \text{round}(s \cdot g_i^t) \\ E(Q(g_i^t)) &= Q(g_i^t) + r_i \end{aligned} \quad (3)$$

where g_i^t is the original gradient; $Q(g_i^t)$ is the quantized gradient; s is the scaling factor used for quantization; $E(Q(g_i^t))$ is the encrypted quantized gradient; r_i is the random mask for client i .

$$g^t = \frac{1}{n} \sum_{i=1}^n Q(g_i^t) + r_i \quad (4)$$

where $g(t)$ is the Aggregated encrypted gradient at round.// This approach resulted in reduced bandwidth usage, faster communication, and smaller update payloads. To evaluate the efficiency, compliance, and viability, FedHealthCare is tested on real healthcare data sets.

- **Real-world deployment and evaluation**—The framework is tested on real healthcare datasets in a simulated environment. This demonstrates feasibility, compliance, and effectiveness in privacy-preserving model training.

3 | Preliminary

A concise mathematical discussion on FL and HE is done in this section. This discussion makes the article self-contained.

3.1 | Federated Learning

FL is a leading-edge decentralized machine learning approach that allows several institutions to collaborate and develop a global model without revealing sensitive information. FL differs from traditional centralized learning, which collects raw data from several institutions in one central place for model training; FL ensures that data always stays local within each institution. Despite sharing raw patient data, each institution trains a local model using its raw data, and then, using a global server, each institution shares only the encrypted model updates or gradients.

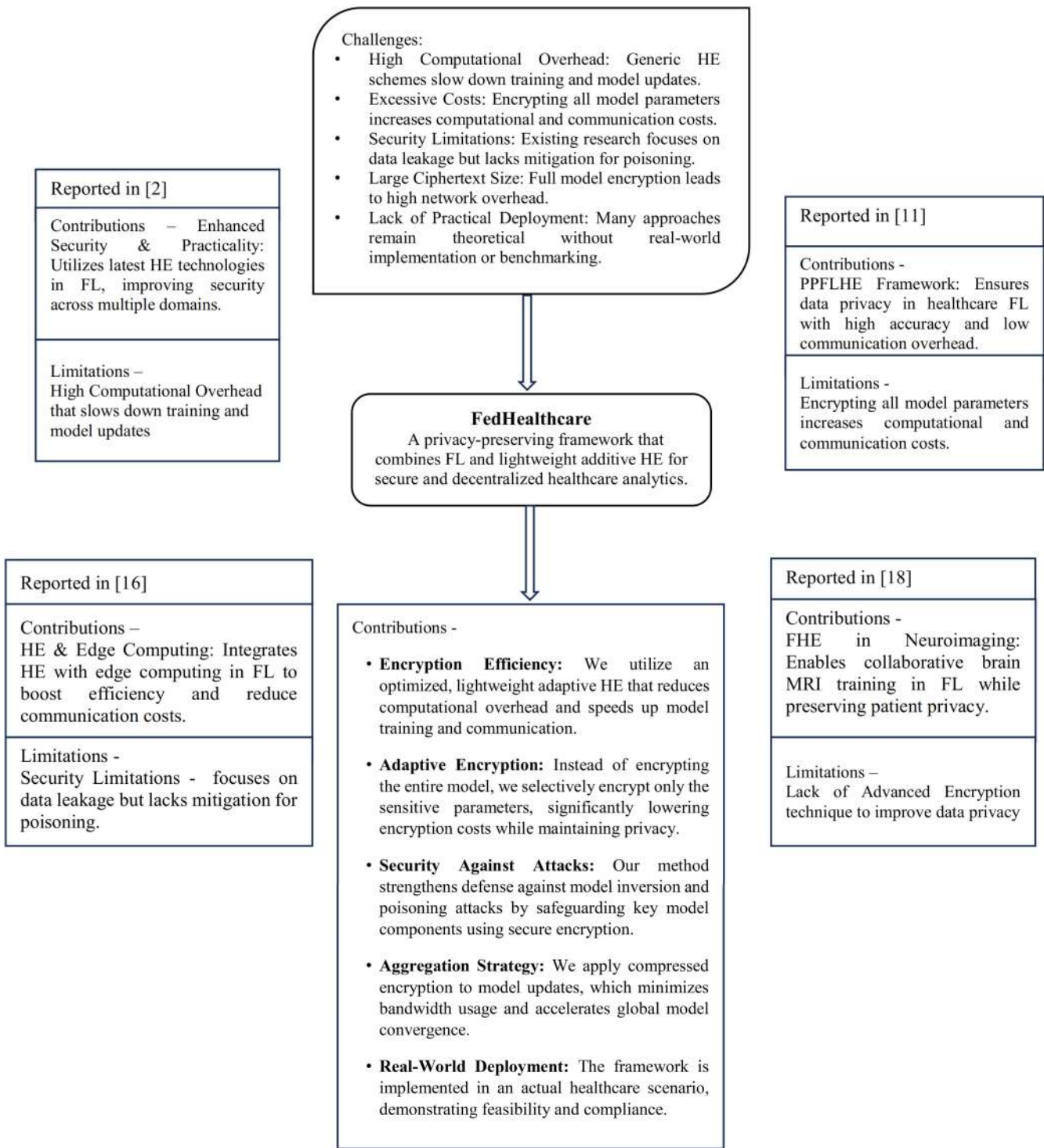


FIGURE 2 | Challenges of schemes reported in [2, 11, 16, 18] and contributions of the FedHealthcare.

These model updates collectively enhance the global model and provide guidance for collaborative learning while obeying the privacy laws like the GDPR and HIPAA. FL is an important step for preserving privacy in healthcare institutions because of the advanced data security needs enforced by these standards. Collaborating institutions will be initialized with a base model developed by the global server, as that is the first step in FL global model initialization. Each institution will then use their own private dataset for local model training, ensuring

that the raw data never leaves its local source, and this helps in learning unique patterns for their facility. Each institution sends their encrypted model updates to the global server only after local training. To enhance the global model, these model updates can be merged together using an approach called Federated Averaging (FedAvg), which computes an average of all received model updates. This iterative process is continued for 10 rounds until the global model integrates into an ideal state.

The global model is initialized as M_0 with random model weights w_0 at the beginning of the FL process. Each institution “ i ” at each round “ t ” produces local updates Δw_i^t by training a local model on its own private dataset D_i . The global server collects these model updates and aggregates all of them into a global model as M_{t+1} . The global model update in FedAvg is written in Equation (5).

$$w_{t+1} = w_t + \frac{1}{N} \sum_{i=1}^N \Delta w_i^t \quad (5)$$

where w_t is the global model weights in round t ; “ N ” is the number of participating hospitals; $\Delta w_i^t = w_i^t - w_t$ represents the local update, with w_i^t are the weights after local training in D_i .

Despite FL’s privacy advantages, it is prone to security threats such as model inversion attacks, in which attackers try to copy or steal private patient data by exploiting the shared model updates. Moreover, FL facilitates frequent interaction between institutions and the global server but can limit the real-time applications because of high bandwidth consumption and increased computation expenses.

3.2 | Homomorphic Encryption

HE helps in improving FL security while constantly securing the patient sensitive raw data. HE allows operations to be performed on the encrypted patient information without the necessity to decrypt them first. HE computes mathematical operations like averaging, addition, and so on, directly on encrypted data values; hence, HE is ideally suited for FL as it can be used to aggregate model updates without sharing sensitive raw data.

Each institution will be training its own local model using homomorphically encrypted patient data. First, the model updates are encrypted using HE and then encrypted model updates are sent to the global server. The global server then aggregates these encrypted model updates without the need to decrypt them; this helps in ensuring that no sensitive raw data is accessible. After the aggregation is complete, the hospitals receive the final encrypted model back and decode it to create the improved global model. This approach considerably lowers the risk of data breaches while maintaining compliance with strict privacy needs. FL combined with HE can facilitate collaborative model training across multiple institutions while securing data privacy.

HE improves FL security by allowing operations to be performed on encrypted model updates without the need to first decode them. In this approach, each institution encrypts its local model update Δw_i^t using a simple HE public key P_k before sending it to the global server. Encryption for a plaintext update $m = \Delta w_i^t$ using a public key P_k is expressed below.

$$c = E(m, P_k) = m + k \quad (6)$$

where “ k ” is a random encryption key unique to each update, ensuring the original value “ m ” is hidden. The server aggregates these encrypted updates homomorphically.

$$E(M_{t+1}, P_k) = E(w_t, P_k) + \sum_{i=1}^N E(\Delta w_i^t, P_k) \quad (7)$$

The additive property of HE states that the server can sum the encrypted updates without decrypting them.

$$E(m_1) + E(m_2) = E(m_1 + m_2) \quad (8)$$

Here, w_t is the current global model and “ N ” is the number of hospitals. The result remains encrypted until the server uses a private key S_k to decrypt it.

$$M_{t+1} = D(E(M_{t+1}, P_k), S_k) = E(M_{t+1}, P_k) - K \quad (9)$$

where “ K ” represents the combined encryption keys removed during decryption. After aggregation, the final model update is decrypted only by authorized hospitals using the private key S_k and it is written in the Equation (10).

$$D(E(w^{t+1}, P_k), S_k) = w^{t+1} \quad (10)$$

where $D(E(x, P_k), S_k)$ is the decryption function using the secret key S_k . This combined process reduces the risk for gradient leaking by securing the model updates during transmission and aggregation of model updates. HE safeguards privacy and supports secured interaction between institutions by restricting the spreading of sensitive raw data. HE is more beneficial than the conventional encryption in preserving privacy in FL. One of the main advantage of HE is its protection against model inversion attacks and even if model updates are intercepted, hackers will not be able to determine any sensitive information from them because they are already being encrypted. Gradient leaking attacks are important security flaws in conventional FL implementations and as a result they are avoided. Moreover, HE allows secure aggregation of model updates without revealing the sensitive data and also offers security, efficiency, and accuracy.

It is already mentioned that the proposed work uses simplified additive HE and it is written in Equation (11)

$$\begin{aligned} c &= E(m, P_k) = m + k \\ \text{or} \\ E(w_i^t) &= w_i^t + K \quad (\text{AdditiveMasking}) \end{aligned} \quad (11)$$

“ k ” is a random mask, significantly reducing ciphertext size and encryption time. Instead of computing:

$$E(w_i^t) = \text{Paillier}(w_i^t) \quad (12)$$

A lot of computing power is needed for this approach. The proposed masked additive approach is sufficient for secure aggregation in FL and provides faster encryption and decryption, reduced bandwidth, and additive HE characteristics without full-blown HE cost. The target shallows the layers in an attempt to mimic model inversion attacks. As a result, only sensitive layers or gradients are encrypted. Thus, it lowers the HE computation’s overall cost. Let, $\theta = (\theta_1, \theta_2, \dots, \theta_n)$ be the all model parameters. It can be encrypted ($\theta_s \subset \theta$) only if it satisfies equation (2).

This leads to the time complexity.

$$T_{enc} = O(\theta_i) \ll O(\theta) \quad (13)$$

4 | Proposed Methodology of FedHealthcare

The main objective of FedHealthcare is to provide a collaboration among the health organizations without compromising patient privacy. The organizations only provide the model parameters rather than sharing the raw medical data. This prevents unwanted access to private information of the patients. This scheme has a decentralized design. Hence, the chance of a single point of failure and security flaws is less. Moreover, the scheme is scalable so more health institutions can join if required. The schematic diagram is shown in Figure 3. It shows the interaction between the health organizations as clients, the encryption module, and the global server in the FL-HE framework.

The implementation of FedHealthcare for privacy-preserving FL using HE in collaborative healthcare data analysis is designed to ensure data privacy, efficient computation, and scalability. It is already mentioned that the proposed approach has developed a lightweight additive HE scheme that encrypts only essential model parameters. It significantly reduces computational and communication overhead. The full implementation is outlined, divided into key modules. Each of them is discussed below.

- **Global model setup:** This step initiates the training process. To initialize the global model M_0 random weights are assigned.
- **Key generation:** In this step HE provides security to the healthcare data. Here, two encryption keys are generated a private key, s_k and a public key, p_k . s_k decrypts the aggregated updates and p_k encrypts the model updates. During system setup, the key generation module produces a shared

p_k and s_k . Each client encrypts its model updates using additive masking.

$$c = m + k \quad (14)$$

where “ k ” is a fresh random mask generated for each communication round using a secure pseudo-random function. The encrypted updates are aggregated via homomorphic addition and the accumulated masks are removed during decryption.

$$D(E(M_{t+1}, p_k)s_k) = E(M_{t+1}) - K \quad (15)$$

where “ K ” is the sum of random masks.

- **Public key distribution:** The public key p_k is shared among the clients. It ensures that they can perform encryption before the transmission to the server.
- **FL process:** It is an iterative process to improve the model updates through client–server interactions. It is done in “ N ” communication rounds.
- For each round “ t ” ($0 < t \leq N$):
 - **Model distribution:** It ensures the receiving of the most recent global model M_t from the global server by every client. This model is the foundation for local training and is essential for coordinating client models toward a shared goal.
 - **Client-side training:** Every client obtains its local dataset, D_i and uses the received model, M_t , to conduct local training. This is calculating the model updates ΔM_i^t by implementing FL algorithm to the local data. It ensures that the model is adjusted with the specific features of the data from every client. The clients use the public key p_k

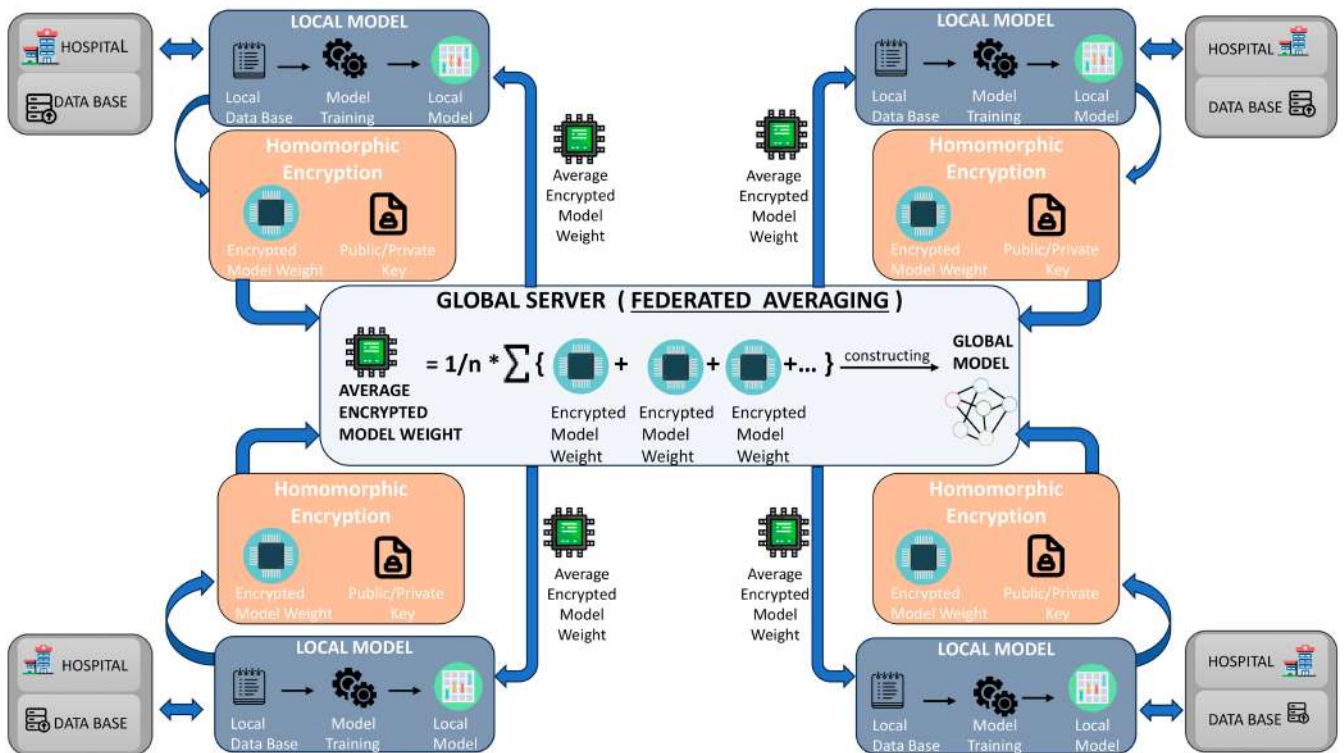


FIGURE 3 | System architecture of FedHealthcare.

ALGORITHM 1 | FedHealthcare.

Input: Healthcare datasets $D_0, D_1, \dots, D_n$ containing patient details and labels for readmission; Initial global model parameters " M_0 "; Number of FL rounds = " N ", public and private encryption keys (" p_k " and " s_k ").

Output: Final global model M_T after " t " rounds of FL, Model updates securely stored on Drive.

```

1: function FedHealthcare( $D_0, D_1, \dots, D_n, M_0, N, p_k, s_k$ )
2:    $M_0 \leftarrow \text{Initialize GlobalModel}()$ —Server Initialization.
3:    $(p_k, s_k) \leftarrow \text{Generate Encryption Keys}()$ —Generating HE keys.
4:   Distribute Key ( $p_k$ )—Sends public key to all participating clients.
5:   for  $t = 1$  to  $N$  do
6:     Send-Model-To-Clients( $M_t$ )—Server sends current global model to all participating clients
7:     Each client performs local training on their dataset, calculating local model updates.
8:     for  $i = 1$  to  $M$  do
9:        $D_i \leftarrow \text{GetClientDataset}(i)$ —Loading and preprocessing Client " $i$ " healthcare dataset " $D$ ".
10:       $\Delta M_i^t \leftarrow \text{Local Training}(D_i, M_t)$ —Compute local model updates using their dataset.
11:       $\text{Encrypted}(\Delta M_i^t) \leftarrow \text{Encrypt}(\Delta M_i^t, p_k)$ —Client " $i$ " encrypts model updates using HE
12:      Server-Side-Aggregation—client " $i$ " sends their encrypted updates to the central server.
13:      Send-Encrypted-Update-To-Server( $i, \Delta M_i^t$ )—(Homomorphic Aggregation)
14:    end for
15:     $\text{Encrypted}(M_{t+1}) \leftarrow \text{Encrypted}(M_t)$ —Start aggregation with current model
16:    for  $i = 1$  to  $N$  do
17:       $\text{Encrypted}(M_{t+1}) \leftarrow \text{Encrypted}(M_t) + \text{Encrypted}\Delta M_i^t$ —Aggregate updates
18:    end for
19:     $M_{t+1} \leftarrow \text{Decrypt}(\text{Encrypted}(M_{t+1}), s_k)$ —Decrypt the aggregated result
20:    Send-Model-To-Clients( $M_T$ )—Send final global model back to clients
21:  end for
22:  Client-Side Verification (AFTER all rounds)
23:  Client decrypts their original update and compares it with the final global model
24:  if  $\Delta M_i^T == \text{Decrypt}(\text{Encrypted}\Delta M_i^T, s_k)$  then
25:    Print ("Privacy Preserved for Client",  $i$ )
26:  else Print ("Privacy Breach Detected for Client",  $i$ ) and then Abort()—if there's a mismatch
27:  end if
28:  Return  $M_T$ —Return the Final global model after  $N$  rounds
29:  Load and store the encrypted model updates on Drive.
30: end function

```

to encrypt the model updates. It produces the encrypted updates (ΔM_i^t). Since the clients never exchange raw updates; HE enables them to protect their data. The global server receives these encrypted updates for aggregation.

- Server-side aggregation:
 - This process is initiated after sharing every client shares the encrypted updates with the global server from every client.
 - * To initialize the aggregation it uses the encrypted global model

$$\text{Encrypted}(M_{t+1}) = \text{Encrypted}(M_t) \quad (16)$$

- * Homomorphic addition is used by the server on the encrypted updates. It can be done without any decryption of the aggregated model. This step is useful to preserve the security of the sensitive data,

$$\text{Encrypted}(M_{t+1}) = \text{Encrypted}(M_{t+1}) + \text{Encrypted}(\Delta M_i^t) \quad (17)$$

- The server decrypts the results after receiving the modifications. The decryption is done using the private key s_k . These are then updated in the global model $M(t+1)$.

- Distribution of final model: The final global model M_T is distributed among each of the clients. This is done only after the completion of all the communication rounds. Every client receives the training effort and enhances the local model accordingly. This is done without sharing any private information of the clients.
- Client-side verification: The trustworthiness in FL environments are maintained in this step. It ensures that there is no inconsistency that would create a privacy breach. The client confirms the privacy if the updates are matched otherwise, the client halts the process and generates potential issues.

At the end of FL process, all clients receive the final global model M_T . It ensures that each client benefits from this collaborative training. The privacy of health data is maintained by iterative enhancement of the performance of individual models. HE has ensured that raw data is not shared throughout the process. It provides a strong foundation for protection of ML privacy. The client-side verification mechanism is maintained by the FL environment. This works as an additional measure to prevent the privacy violations throughout the updation process. The corresponding methodology is outlined in Algorithm 1.

5 | Performance Evaluation of FedHealthcare

In this section, the experimental findings and the comparative analysis of FedHealthcare are reported. The proposed framework integrates HE with FL for safe healthcare data analysis in different health organizations. The findings confirm that FL and HE work well together to protect privacy while preserving high model performance, which makes it a good option for extensive medical AI applications. The results of FedHealthcare are interpreted in terms of accuracy, security robustness, communication overhead, computing cost, experimental performance, and comparison with some recent related works.

5.1 | Experimental Setup and Dataset

FedHealthcare is tested using a real-world medical dataset comprising extensive patient records that included various clinical, diagnostic, and demographic data. The experimental evaluation was conducted on Google Colab, with libraries such as

Scikit-learn, PySyft, and TensorFlow being utilized. A FL system was simulated with four clients, each representing a distinct healthcare facility and being provided access to a substantial number of patient records from their respective datasets.

5.1.1 | Dataset Description

Each client's dataset contains substantial patient data covering demographics (age, gender, and race), admission and discharge details, ward and doctor assignments, and diagnosis information. The features used for modeling, after preprocessing, include patient age, encoded gender, wait time, length of stay, and encoded values for hospital ward, doctor specialty, and diagnosed disease. Each client runs on a separate, independent dataset of roughly 10 000 records, organized as if each one were utilized to determine the readmission rate of the patient in actual practice [32, 33]. Only encrypted model changes are sent to a centralized server using the privacy-preserving Federated Averaging protocol. There is no need for sharing raw data. Table 1 shows the records of readmission of patients. Here, a binary label with "1" represents the readmission of the patients, and "0" denotes that they were not. It is the target variable.

5.1.2 | Correlation Matrix

A correlation heat map is shown in Figure 4. It represents the assessment of multicollinearity and finds pertinent correlations between variables. Patient length of stay represents a substantial

TABLE 1 | Client-wise distribution of hospital readmission data.

Client ID	Records	Class balance
Client-1	≈10 000	0 → ≈59% • 1 → ≈41%
Client-2	≈10 000	0 → ≈58% • 1 → ≈42%
Client-3	≈10 000	0 → ≈60% • 1 → ≈40%
Client-4	≈10 000	0 → ≈59% • 1 → ≈41%

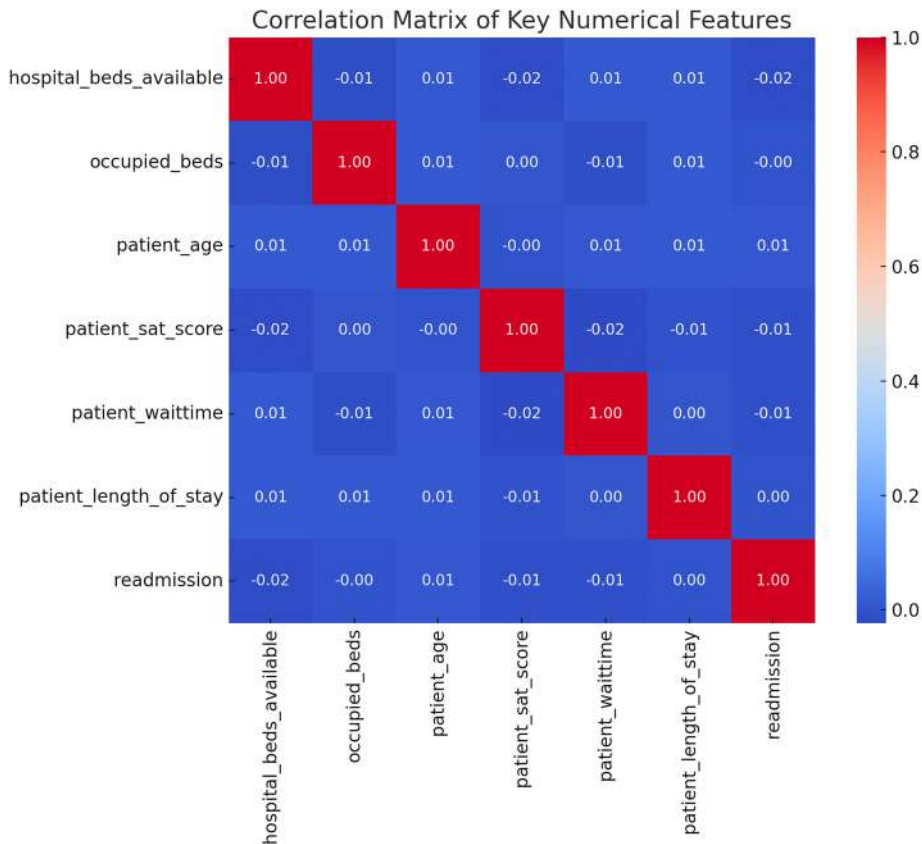


FIGURE 4 | Correlation matrix for key numerical features.

positive association. As per this matrix, longer wait times are frequently associated with longer stays in hospitals or health institutes. It is also observed that patients who were admitted for a longer period of time may be likely to be readmitted.

5.2 | Computational Cost

To evaluate the computational performance of FedHealthcare, it is compared with a traditional centralized training setup. The primary metric used was training time per round, averaged over “10” communication rounds on a simulated multi-client environment using TensorFlow and PySyft on Google Colab. Table 2 shows the average and total time of the centralized and FL method. The overall training duration increased by almost 20% as a result of federated training, which can be seen in Figure 5. However, considering the privacy benefits, the overhead is justified. Considering the enhanced privacy guarantees offered by FL, where no raw data is transmitted and client data remains local, this trade-off is generally acceptable.

5.3 | Communication Overhead

Communication cost is a vital consideration in FL systems. Specifically, it works with environments with limited bandwidth or large-scale deployments with numerous clients. The number of model updates is sent back and forth between clients and the central server during each FL cycle in order to calculate the communication cost. We specifically assessed the number of clients that participated at certain important training

rounds (i.e., 10, 20, 30, 40, and 50) and the total amount of data transmitted in kilobytes (KB). The results are summarized in Table 3. The communication overhead for each round consists of broadcasting the modified global model back to the clients and sending encrypted model updates from all participating clients to the server. Differences in model update size, which can be impacted by the amount of local training data and adaptive encryption techniques, are the cause of slight differences in the data transferred. Figure 6 shows the communication cost across different numbers of rounds. Even while client participation varied slightly, the communication overhead stayed largely constant at an average of 250 KB every round. This illustrates the effectiveness and scalability of FedHealthcare. It minimizes communication payload by using selective encryption techniques and compressed encrypted updates. This framework achieves a better trade-off between communication cost and data security than techniques that use uncompressed gradients or full HE.

5.4 | Security Analysis

The fundamental tenets of the FedHealthcare framework are security and privacy. Adversarial evaluations aimed at reconstruction assaults and membership inference attacks are carried out. These are two of the most prevalent and worrisome risks in collaborative healthcare systems. To gauge the system’s resistance to privacy breaches, these are done. To protect all the

TABLE 2 | Training time comparison.

Model type	Avg time per round (s)	Total time (10 rounds)
Centralized	1.23	61.5
Federated (4 clients)	1.65	72.5

TABLE 3 | Communication cost matrices.

Number of rounds	Total data exchanged (KB)	Clients participated
10	250	4
20	245	5
30	260	4
40	248	5
50	250	4

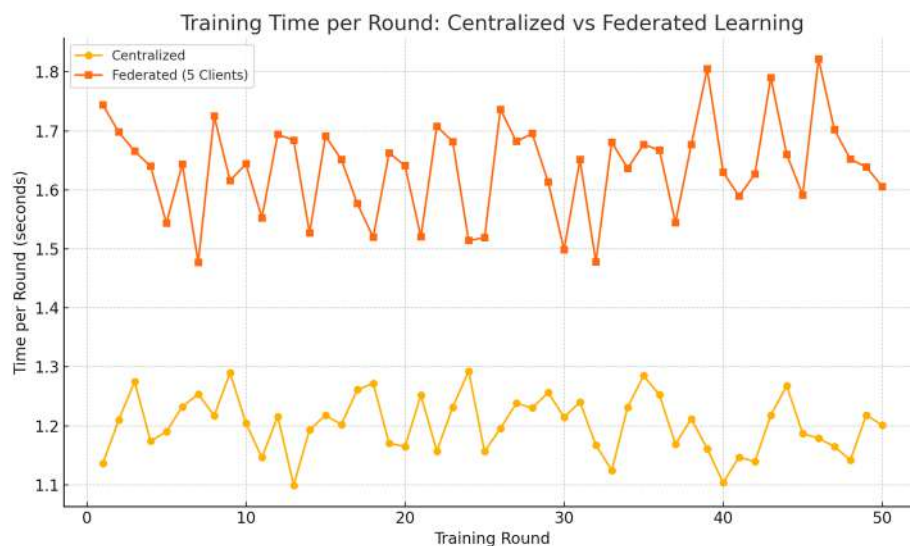


FIGURE 5 | Comparison of training time between centralized and federated learning.

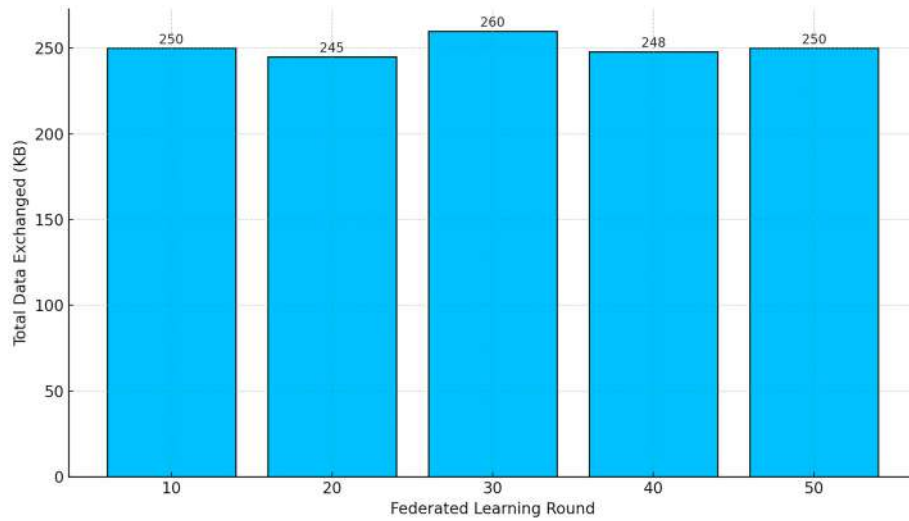


FIGURE 6 | Communication cost across different number of rounds.

TABLE 4 | Security analysis using HE.

Attack type	Without HE (%)	With HE (%)
Reconstruction rate	72	28
Membership inference	81	35

model updates while they are being transmitted and aggregated, the system uses HE. This greatly reduces the surface area for data leakage by guaranteeing that the central server can aggregate encrypted updates without ever having access to raw model parameters. It is listed in Table 4. The adversarial efficacy significantly decreased when the model update sharing mechanism was subjected to HE. While membership inference assaults decreased from 81% to 35%, the rebuilding rate decreased from 72% to 28%. These findings confirm that our framework delivers high privacy guarantees without sacrificing model accuracy, in contrast to approaches that rely on DP. This may involve utility loss owing to noise. Hence, the integrity of the global model is preserved. It allows the safe aggregation of weights and encrypted gradients.

5.5 | Accuracy and Efficiency

FL model securely aggregates the encrypted model updates and may achieve high accuracy for healthcare data. Not only can high prediction values be achieved using FL models, but data privacy can also be maintained in comparison to the centralized models with complete data access. In spite of the decentralized structure, it is mention-worthy that FL models have strong predictive capability. It is very suitable for the environments where privacy is a big concern. The improvement in predictive accuracy over the centralized structure is graphically represented in Figure 7. This line graph demonstrates the performance of the global model in multiple FL training rounds. The computational burden that encryption introduces is one of the main obstacles in integrating HE with FL. This system optimizes HE for efficiency, ensuring FL-HE maintains strong privacy protection with minimal impact

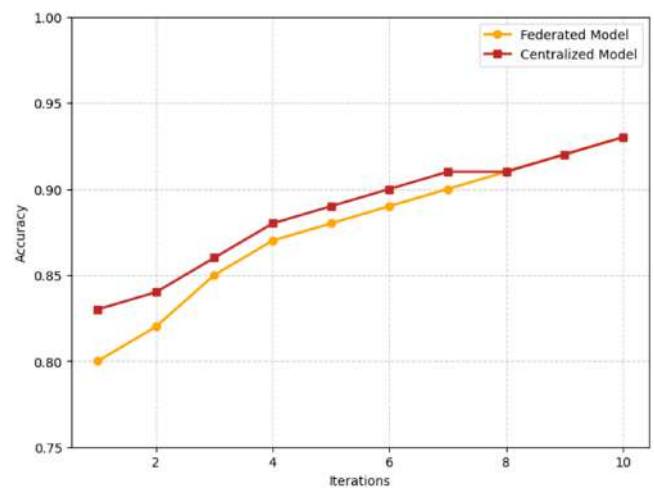


FIGURE 7 | Predictive accuracy over iterations.

on computational performance (Figure 7). This solution ensures real-time, secure processing of healthcare data with an encryption overhead of only about 5%. The trade-off between computational time and accuracy is represented in Figure 8. This graphical representation proves the efficiency of the integrated FL-HE models over the conventional FL models.

5.6 | Comparative Study and Analysis

A comparison study of FedHealthcare with the other related approaches is listed in Table 5. This study is based on various privacy-preserving approaches such as DP [5], Traditional FL [25], DP with FL [7], secure multi-party computation (SMPC) [14] and FL with DP and Blockchain [21]. Table 5 includes the variation in the working principles for the techniques used with the involvement of different parameters. The methods include diverse issues and in that sense do not represent a comparison in the true sense. However, a relative merits and demerits of the above-mentioned approaches are included to highlight the status of the related works.

FedHealthcare strikes a balance between accuracy, security, and computational efficiency, outperforming conventional FL and other privacy-preserving strategies. It incorporates HE to secure model updates without sacrificing performance, in contrast to standard FL, which lacks robust encryption. While adding noise to ensure privacy, DP can decrease accuracy, and SMPC, despite being extremely secure, has enormous computing requirements that make it unfeasible for large-scale FL. On the other hand, it is perfect for real-time, privacy-preserving healthcare applications since it delivers near-centralized accuracy with no computing overhead.

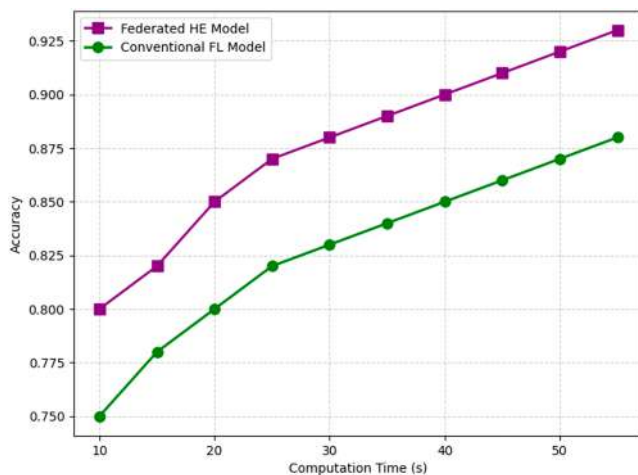


FIGURE 8 | Predictive accuracy over computational time.

A comparative analysis is also performed against some privacy-preserving learning frameworks documented in recent literature such as [5, 11, 17, 22, 30]. These schemes integrate encryption-based or DP mechanisms. To contextualize the efficacy of FedHealthcare, the comparative analysis is listed in Table 6. Accuracy, efficiency, and privacy are three important dimensions that our framework exhibits a solid balance. Although [11] uses HE as well, their method has a larger computational and transmission overhead because it uses full parameter encryption and more complicated encryption techniques. In contrast, FedHealthcare employs gradient compression and selective HE, which makes it more feasible for real-world applications without compromising privacy. Our approach preserves the signal quality by not introducing noise into the model updates, which leads to improved accuracy when compared to DP-only systems.

It is worthy to mention that FedHealthcare proposed a lightweight HE (LWHE) method over conventional HE in FL. Hence, a comparative analysis is done on three crucial parameters, namely “accuracy,” “computing time,” and “communication overhead.” Table 7 lists the FL performance comparison: Normal HE versus the proposed one. FedHealthcare removes noise and encrypts the most sensitive gradients. Hence, the accuracy increases by 5.7%. The proposed methodology uses an optimized encryption and decryption methodology. Hence, there is a significant reduction in the computational cost. Moreover, the communication overhead is reduced from 350 to 250 KB. This is possible by using selective encryption techniques and gradient compression. The encrypted model updates prevent

TABLE 5 | Comparative study with related frameworks.

Approach	Objective	Methodology	Primary contribution
Traditional FL [25]	Learning collaboratively using decentralized data	Aggregates raw updates and shares model updates on the server	Achieves reasonable model accuracy, but lacks strong data encryption, putting privacy at risk in collaborative healthcare
DP [5]	Protection of privacy with statistical assurances	Introduces noise into model updates	Obscures important patterns to protect privacy, although noise addition may lower model accuracy, particularly in smaller datasets
FL with DP [7]	Privacy-preserving, scalable and robust FL for healthcare	Combines DP for obfuscation, HE for secure aggregation, adaptive node weighting and quantized encryption	Complies with HIPAA/GDPR, improves scalability, resists non-IID issues, supports real-time updates
SMPC [14]	Cooperative calculations that maintain high privacy	Data is encrypted for calculations	It is less effective for large-scale or real-time analysis due to its high processing costs, despite its strong privacy assurances
FL with DP and Blockchain [21]	Quantum-secure, privacy-preserving FL for rare diseases	FL with adaptive DP, AES hybrid encryption and blockchain-based verification with IPFS storage	Resists quantum, inference and poisoning attacks; optimizes privacy–utility balance with robust aggregation
FedHealthcare	Decentralized analysis of medical data with improved privacy	Integration of HE and FL to update models securely	Achieves near-centralized model performance with a lower computing load, striking a balance between high accuracy and privacy, making it perfect for scalable healthcare applications

TABLE 6 | Efficiency comparison study of FedHealthcare with respect to the schemes reported in [5, 11, 17, 22, 28, 30].

Scheme	Accuracy	Privacy mechanism	Computation cost	Computation overhead
[22]	85.7	Schnorr-based RFID protocol	Low	Low
[11]	86.1%	HE	High	High
[17]	85.4%	DP	Moderate	Moderate
[5]	84.3%	None	Low	None
[28]	88.6	Full HE	High	High
FedHealthcare	90.8	HE	Moderate	Moderate

TABLE 7 | FL performance comparison—traditional HE versus FedHealthcare.

Parameters	FL with	
	traditional HE [8]	FedHealthcare
Model accuracy (%)	85.1	90.8
Avg time per round (s)	2.45	1.65
Encryption overhead (%)	18%	5%
Communication payload (KB)	350	250

various attacks like model inversion, membership inference, gradient leakage, and so on. The comparative analysis listed in Table 7 proves the effectiveness and ensures the potential of FedHealthcare over traditional HE application reported in [8].

6 | Conclusion and Scope of Future Work

FedHealthcare proposes an integrated framework of FL and lightweight additive HE in the field of privacy-preserving medical data analytics. The framework facilitates secure, decentralized learning across multiple healthcare institutions. It enables collaborative FL model training without compromising patient privacy. Health institutes use patient data and train ML models locally without sharing raw data. In this scheme, lightweight additive HE encrypts only the essential model parameters. FedHealthcare achieves 90.8% accuracy over 10 rounds across four clients, reducing communication cost by 29%, encryption overhead from 18% to 5%, and training time from 2.45 to 1.65 s over the scheme reported in [8]. Despite the efficiency, challenges remain with communication overhead and cipher-text expansion in HE. The FedHealthcare framework can offer strong potential for applications in healthcare. However, it does not address active adversarial scenarios, including model poisoning and Byzantine attacks. These may compromise the integrity of the model during training. To minimize this, it may be considered as the scope of future research. Additionally, the system's reliance on a centralized aggregator creates a single point of failure, while HE still causes cipher-text expansion and overhead. It affects scalability in resource-limited settings. Hence future work may also focus on

blockchain-based decentralized aggregation, Byzantine-resilient protocols, and integration with SMPC to enhance the robustness and scalability of the framework.

Author Contributions

All the authors have contributed equally in preparation of the manuscript.

Ethics Statement

This article does not contain any studies with human participants or animals performed by any of the authors.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

1. M. Babar, B. Qureshi, and A. Koubaa, "Review on Federated Learning for Digital Transformation in Healthcare Through Big Data Analytics," *Future Generation Computer Systems* 160 (2024): 14–28.
2. G. Shen, Z. Fu, Y. Gui, W. Susilo, and M. Zhang, "Efficient and Privacy-Preserving Online Diagnosis Scheme Based on Federated Learning in e-Healthcare System," *Information Sciences* 647 (2023): 119261.
3. H. R. Kumbhar and S. Srinivasa Rao, "Federated Learning Enabled Multi-Key Homomorphic Encryption," *Expert Systems with Applications* 268 (2025): 126197.
4. M. Firdaus, H. T. Larasati, and K. Hyune-Rhee, "Blockchain-Based Federated Learning With Homomorphic Encryption for Privacy-Preserving Healthcare Data Sharing," *Internet of Things* 31 (2025): 101579.
5. G. Sathish Kumar, K. Premalatha, G. Uma Maheshwari, P. Rajesh Kanna, G. Vijaya, and M. Nivaashini, "Differential Privacy Scheme Using Laplace Mechanism and Statistical Method Computation in Deep Neural Network for Privacy Preservation," *Engineering Applications of Artificial Intelligence* 128 (2024): 107399.
6. B. M. Gandhi, S. B. Vaghadia, M. Kumhar, et al., "Homomorphic Encryption and Collaborative Machine Learning for Secure Healthcare Analytics," *Security and Privacy* 8, no. 1 (2025): e460.
7. A. Ali, V. Snášel, and J. Platoš, "Health-FedNet: A Privacy-Preserving Federated Learning Framework for Scalable and Secure Healthcare Analytics," *Results in Engineering* 27 (2025): 106484.

8. D. Truhn, S. Tayebi Arasteh, O. L. Saldanha, et al., "Encrypted Federated Learning for Secure Decentralized Collaboration in Cancer Image Analysis," *Medical Image Analysis* 92 (2024): 103059.
9. F. Cremonesi, V. Planat, V. Kalokyri, et al., "The Need for Multimodal Health Data Modeling: A Practical Approach for a Federated-Learning Healthcare Platform," *Journal of Biomedical Informatics* 141 (2023): 104338.
10. D. Stripelis, H. Saleem, T. Ghai, et al., "Secure Neuroimaging Analysis Using Federated Learning With Homomorphic Encryption," in *17th International Symposium on Medical Information Processing and Analysis*, vol. 12088 (SPIE, 2021), 1208814.
11. B. Wang, H. Li, Y. Guo, and J. Wang, "PPFLHE: A Privacy-Preserving Federated Learning Scheme With Homomorphic Encryption for Healthcare Data," *Applied Soft Computing* 146 (2023): 110677.
12. S. Kanchan, J. W. Jang, J. Y. Yoon, and B. J. Choi, "Efficient and Privacy-Preserving Group Signature for Federated Learning," *Future Generation Computer Systems* 147 (2023): 93–106.
13. T. T. Kuo and A. Pham, "Detecting Model Misconducts in Decentralized Healthcare Federated Learning," *International Journal of Medical Informatics* 158 (2022): 104658.
14. J. Qiao, N. Wang, J. Fu, L. Deng, J. Wang, and J. Liu, "A Lightweight CP-ABE Scheme for EHR Over Cloud Based on Blockchain and Secure Multi-Party Computation," *Transactions on Emerging Telecommunications Technologies* 36, no. 2 (2025): e70053.
15. R. U. Z. Wani, F. Thabit, and O. Can, "Security and Privacy Challenges, Issues, and Enhancing Techniques for Internet of Medical Things: A Systematic Review," *Security and Privacy* 7, no. 5 (2024): e409.
16. W. Shafik, *Chapter 1—Digital Healthcare Systems in a Federated Learning Perspective* (Federated Learning for Digital Healthcare Systems Academic Press, 2024).
17. K. Meduri, G. Nadella, A. Yadulla, et al., "Leveraging Federated Learning for Privacy-Preserving Analysis of Multi-Institutional Electronic Health Records in Rare Disease Research," *Journal of Economy and Technology* 11 (2024): 3.
18. D. Stripelis, H. Saleem, T. Ghai, et al., "Secure Neuroimaging Analysis Using Federated Learning With Homomorphic Encryption," *CoRR* (2021) abs/2108.03437.
19. M. Firdaus and K. H. Rhee, "Towards Trustworthy Collaborative Healthcare Data Sharing," in *International Conference on Bioinformatics and Biomedicine (BIBM)* (IEEE, 2023), 4059–4064.
20. Y. Zhou, L. Song, Y. Liu, et al., "A Privacy-Preserving Logistic Regression-Based Diagnosis Scheme for Digital Healthcare," *Future Generation Computer Systems* 144 (2023): 63–73.
21. B. B. Sezer and H. Turkmen, "PPFLQB: A Privacy-Preserving Federated Learning Enhanced Quantum-Secure Blockchain Layered Framework," *Internet of Things* 33 (2025): 101719.
22. M. Shariq, K. Singh, M. Y. Bajuri, A. A. Pantelous, A. Ahmadian, and M. Salimi, "A Secure and Reliable RFID Authentication Protocol Using Digital Schnorr Cryptosystem for IoT-Enabled Healthcare in COVID-19 Scenario," *Sustainable Cities and Society* 75 (2021): 103354.
23. M. Shariq and K. Singh, "A Secure and Lightweight RFID-Enabled Protocol for IoT Healthcare Environment: A Vector Space Based Approach," *Wireless Personal Communications* 127, no. 4 (2022): 3467–3491.
24. A. Kumar, K. Singh, M. Shariq, et al., "An Efficient and Reliable Ultra-lightweight RFID Authentication Scheme for Healthcare Systems," *Computer Communications* 205 (2023): 147–157.
25. D. O. Ogundipe, "The Impact of Big Data on Healthcare Product Development: A Theoretical and Analytical Review," *International Medical Science Research Journal* 4, no. 3 (2024): 341–360.
26. Y. Lu, X. Huang, Y. Dai, S. Maharjan, and Y. Zhang, "Blockchain and Federated Learning for Privacy-Preserved Data Sharing in Industrial IoT," *IEEE Transactions on Industrial Informatics* 16, no. 6 (2019): 4177–4186.
27. S. Panda, R. Dubey, B. Jena, V. Pareek, L. W. Tsai, and S. Saxena, "Federated Learning Frameworks in Smart e-Healthcare: A Systematic Literature Review With Bias Evaluation," *Applied Soft Computing* 171 (2025): 112747.
28. Y. Su, X. A. Wang, W. Du, Y. Ge, K. Zhao, and M. Lv, "A Secure Data Fitting Scheme Based on CKKS Homomorphic Encryption for Medical IoT," *Journal of High Speed Networks* 29, no. 1 (2023): 41–56.
29. R. S. Antunes, C. André da Costa, A. Küderle, I. A. Yari, and B. Eskofier, "Federated Learning for Healthcare: Systematic Review and Architecture Proposal," *ACM Transactions on Intelligent Systems and Technology* 13, no. 4 (2022): 1–23.
30. W. Jin, Y. Yao, S. Han, et al., "FedML-HE: An Efficient Homomorphic-Encryption-Based Privacy-Preserving Federated Learning System," *arXiv Preprint arXiv:230310837* (2023).
31. H. Ku, W. Susilo, Y. Zhang, W. Liu, and M. Zhang, "Privacy-Preserving Federated Learning in Medical Diagnosis With Homomorphic Re-Encryption," *Computer Standards & Interfaces* 80 (2022): 103583.
32. UCI Machine Learning Repository, "Early Stage Diabetes Risk Prediction," <https://doi.org/10.24432/C5VG8H>.
33. UCI Machine Learning Repository, "Heart Disease," <https://doi.org/10.24432/C52P4X>.